EÖTVÖS LORÁND UNIVERSITY
FACULTY OF INFORMATICS

Thesis Registration Form

Attention! The deadline for amendment is 1 February in case of final exam in June and 31 August in case of final exam in January.

Student's Data:
Student's Name: Kanybekov Nurmuhammed
Student's Neptun code: NRTKMF

Educational Information:
Training programme: Computer Science BSc
I have an internal supervisor

Internal Supervisor's Name: *Walid Guettala*
Supervisor's Home Institution: ***Department of Artificial Intelligence***
Address of Supervisor's Home Institution: ***1117, Budapest, Pázmány Péter sétány 1/C.***

Supervisor's Position and Degree: *PhD candidate, MSc*

Thesis Title: Design and Implementation of AkVen Integrated E-Commerce and Inventory Management System

Topic of the Thesis:
(Upon consulting with your supervisor, give a 150-300-word-long synopsis os your planned thesis. Please do not use the "&" character in the text, otherwise the request will not generate correctly.)

Motivation
AkVen Socks is our family business specializing in the import and distribution of high quality socks from the renowned NabiYa manufacturer. Currently, stock tracking, order processing, and sales analysis are handled manually or with basic tools. This leads to frequent inventory discrepancies, delayed order fulfillment, and insufficient visibility into business performance.
I plan to design and implement a custom full stack web application for my family business, AkVen Socks, the official distributor of premium NabiYa socks. The system will consist of a modern customer facing ecommerce website and a secure administrative dashboard for managing products, inventory, orders, and sales. The application will be developed from scratch following a strict three layer architecture consisting of frontend, backend with business logic, and database layers.

Project Idea
The thesis will involve the development of a complete custom ecommerce and inventory management system. The application will include two main components:
A modern online store with a clean, visually appealing design and intuitive user interface where customers can easily browse, filter and buy socks
A secure administrative dashboard for managing the entire business operations
The system will be built entirely from scratch without using any content management platforms such as WordPress or Shopify. The primary objective is to replace the existing manual processes with an efficient, scalable, and maintainable software solution.

Main Features
Customer facing website:
Product catalog with advanced filtering by color, size, price, collection and other attributes
Detailed product pages with multiple images and variant selection
User registration, login, and personal order history
Shopping cart and checkout functionality

Admin dashboard:
Full CRUD operations for products including images and variants
Real time inventory management with automatic low stock alerts
Order management including viewing and status updates
Simple sales analytics with visual charts such as best selling items and revenue trends

Additional technical features:
Fully responsive design for both desktop and mobile devices
Secure authentication and role based access control

Technical Architecture:
The system will follow the classic three layer architecture:
Frontend layer using a modern frontend framework
Business logic layer containing core application rules for orders, inventory and alerts

Database layer with a well designed relational database and normalized schema

Budapest, 2026. 10. 10.